Chapter 1: Coupled Simulation

In this chapter, we couple the locally mass-conserving MFEM and FVM methods to perform numerical simulations of a partially molten mantle, with the remaining uncertainty in porosity ϕ addressed through phase package evaluation. We describe the implementation details and coupling strategy that integrate the MFEM, FVM, and the eutectic phase package. Before tackling the fully coupled problem, we first examine an alternative porosity evolution scenario based on a prescribed melting rate. We then present an example that combines the MFEM, FVM, and eutectic phase package into a fully coupled computation. The discussion begins with pseudo-1D scenarios and subsequently extends to full 2D simulations.

1.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ denote the cell-averaged solution to the transport system (**??**) at time t^n. Similarly, let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ represent the solution to the Darcy-Stokes system at the same time t^n. Finally, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n obtained from the transport result $\mathcal{U}^n$.

For each time step the time t^n, we begin by reconstructing the dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$ using ML-WENO reconstruction at each data point of interest. The eutectic phase package then provides the corresponding phase parameters $\mathcal{W}^n$ based on these point wise values. With $\mathcal{W}^n$ determinedin particular with ϕ^n, we solve the Darcy-Stokes system at the fixed time t^n to obtain $\mathcal{V}^n$. Finally, the cell-averaged solution $\mathcal{U}^{n+1}$ at the next time t^{n+1} is then obtained by forward marching of the transport system. The complete procedure is summarized as a coupled solver in Algorithm 1.

Algorithm 1 Sequential Coupled Solver

Let $\mathcal{U}^0$ be given as initial data.

1: **for** Time level $n = 0, 1, 2, \ldots, n_{\max} - 1$ **do**

2: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.

3: Using ϕ^n in W^n, solve the Darcy-Stokes system (**??**)–(**??**) for $\mathcal{V}^n$.

4: Given $\mathcal{W}^n$ and $\mathcal{V}^n$, compute dimensionless effective velocity (**??**) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi \check{\mathbf{v}}_l + D_i(1-\phi)\check{\mathbf{v}}_s}{\phi + (1-\phi)D_i}, \quad \check{\bar{\mathbf{v}}} = \phi \check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

5: Marching forward the transport system (**??**) to the next time $n+1$ for $\mathcal{U}^{n+1}$.

6: **end for**

1.1.1 Discontinuous Porosity

Due to the presence of no melt regions and distinct eutectic phase regions, the reconstructed values of the dimensionless enthalpy and mass concentration may differ on the two sides of an edge shared by adjacent elements. This, in turn, can lead to two different porosity values being computed for the same edge. However, in a conforming MFEM scheme, it is essential to assign a single consistent and physically appropriate porosity value on each element edge.

Therefore, we evaluate the harmonic average of the two values of porosity ϕ_e^+ and ϕ_e^- at each quadrature points on both sides of an element edge e to obtain

$$\overline{\phi}_e = \frac{2\phi_e^+ \phi_e^-}{\phi_e^+ + \phi_e^-}. \tag{1.1}$$

The harmonically averaged porosity $\overline{\phi}_e$ is then used in the continuous compaction equation (**??**), where the face integral is converted to an edge integral via the divergence theorem.

Importantly, when the porosity on one side of the edge vanishes, the geometric average also becomes zero. This property enforces the physical condition that no liquid can pass directly through an edge adjacent to a no-melt region.

1.2 Pseudo One-Dimensional Column Tests

In this section, we test the coupled algorithm on a two-dimensional $M \times N$ grid with a pseudo one-dimensional compaction column problem. We study a compacting column aligned along the y-direction, defined over the domain $y \in [-H, 0]$ and $x \in [-L, L]$. Two scenarios are investigated.

- Prescribed melting case: The MFEM and FVM solvers are coupled without phase computation. Instead, a simple prescribed melting function $\Gamma(\mathbf{x})$ is used.
- Phase calculation case: Phase transition and porosity evolution are computed simultaneously, providing a complete coupling between the MFEM, FVM, and eutectic phase package.

For the two tests presented below, we take domain dimensions $H = 2$ and $L = 0.1$. We fix mesh resolution in the x-direction as $M = 2$ and pick different resolutions in y-direction. We fix the parameter $\Theta = 0$ as well. The code is implemented in PETSc/C++, with implementation details given in the appendix.

1.2.1 Prescribed Melting Cases

Before coupling the mechanical, transport and phase packages together into one integrated computation, we first illustrate a simpler model that couples MFEM and FVM solvers without involving the phase package. In this reduced setting, the porosity distribution is not determined by phase-splitting calculations. Instead, we prescribe the melting rate or, equivalently, the porosity evolution through a fixed source function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \tag{1.2}$$

We express conservation of liquid mass by introducing the prescribed melting term $\Gamma(\mathbf{x})$ as a source term on the right-hand side of the advection equation as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \tag{1.3}$$

where the diffusion term is being dropped for simplicity in this case.

We prescribe the following boundary conditions for the MFEM and the FVM solvers respectively to replicate the behavior of the one-dimensional problem within the two-dimensional grid.

Consider first the Darcy-Stokes MFEM solver. On the bottom edge $y = -H$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with an essential boundary conditions with a constant upwelling velocity v, so

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e v\, ds;$$

- Tangential Stokes velocity is prescribed with an essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with an essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

On the top edge $y = 0$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with natural boundary conditions, i.e., traction free,

$$t_0 = 0;$$

- Tangential Stokes velocity is prescribed with a zero essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with a natural boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

On the left and right Edges $x = \pm L$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = 0;$$

- Tangential Stokes velocity is prescribed with the natural boundary condition of free traction,

$$t_0 = 0;$$

- Normal Darcy flux is prescribed with a non-permeable essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

Finally, for the transport FVM solver, on the bottom edge $y = -H$, impose that:

- A corresponding Dirichlet boundary condition is prescribed on the inflow boundary aligning with the essential boundary condition specified in the MFEM solver.

On the top edge $y = 0$, impose that:

- A free outflow boundary condition is imposed, consistent with the free traction boundary condition applied in the MFEM solver.

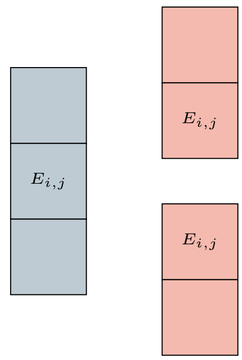

Figure 1.1: An exhibition of a set of 1D ML-WENO (3,2) stencils on a rectagular mesh.

On the left and right edges $x = \pm L$, impose that:

- A no-flow boundary condition is prescribed consistent with the zero-flow Dirichlet boundary used in the MFEM solver.

For simplicity and efficiency in implementation and computation, we employ one-dimensional ML-WENO(3,2) stencils as illustrated in Figure 1.1, since higher-order accuracy in the x-direction is unnecessary for this problem. We use second order SSP Runge-Kutta time integration scheme defined in Equation (**??**).

We consider two different scenarios, each characterized by distinct prescribed melting functions and initial conditions:

Case 1: A piece-wise constant source function.

$$\Gamma_1(z) = \begin{cases} 2.5e-3, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \tag{1.4}$$

We assign an initial condition of no melting everywhere as

$$\phi_1(z,0) = 0.0, \quad z \in [-2.0, 0.0]. \tag{1.5}$$

Figure 1.2 illustrates the temporal evolution of porosity, Stokes and Darcy velocities, Stokes and Darcy pressures, as well as the effective pressure, defined as $p_{\text{eff}} = p_s - p_l$.

The third panel, corresponding to dimensionless time $\check{t} = 1325$, demonstrates that the system has reached a steady state after sufficient evolution.

Case 2: We next consider a slightly more complicated scenario in which a melting source is trapped beneath the bottom of an existing porosity channel. The trapped melting source produces a step in porosity, thereby induces an instantaneous perturbation in the compaction rate. The source function is

$$\Gamma_2(z) = \begin{cases} 0, & -1.5 < z \leq 0, \\ 1.5e-3, & -1.8 < z \leq -1.5, \\ 0, & -2.0 \leq z \leq -1.8, \end{cases} \tag{1.6}$$

where we assign an initial condition with an existing porosity channel as

$$\phi_2(z, 0) = \begin{cases} 5e-2, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \tag{1.7}$$

Figure 1.3, 1.4 and 1.5 illustrate the temporal evolution of porosity, Stokes and Darcy velocities and Stokes and Darcy pressures. A propagating discontinuity in porosity forms when the porosity profile decreased upward (See Katz (2022)). Eventually, the system reaches a steady state, characterized by a narrower porosity channel. The simulation demonstrates robust and stable evolution, despite the presence of a highly complicated porosity profile. The solution preserves its overall structure under mesh refinement.

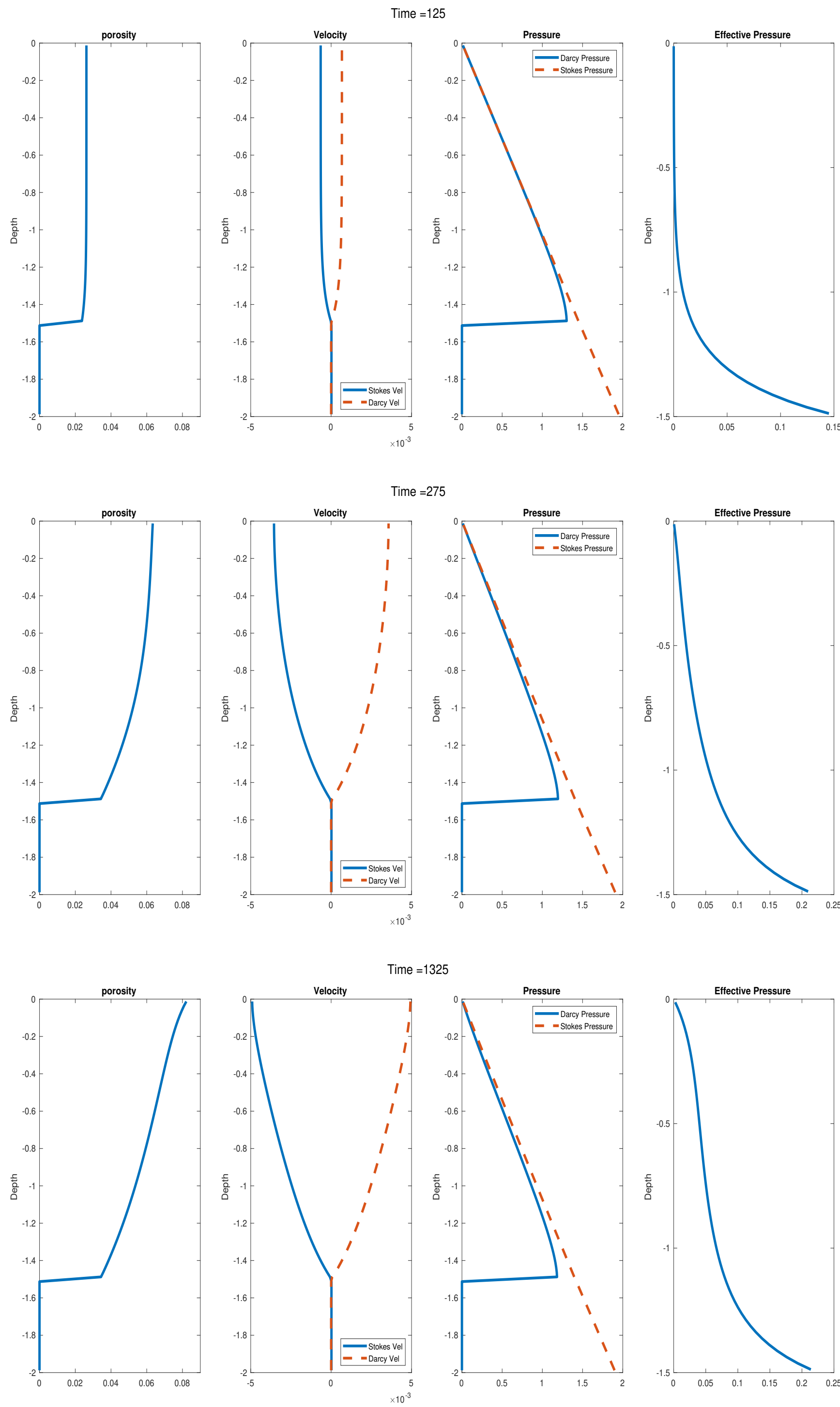

Figure 1.2: Prescribed melting case with constant melting source term 1.4. Shown at time = 125, 275 and 1325, where the system reaches steady state at the time = 1325.

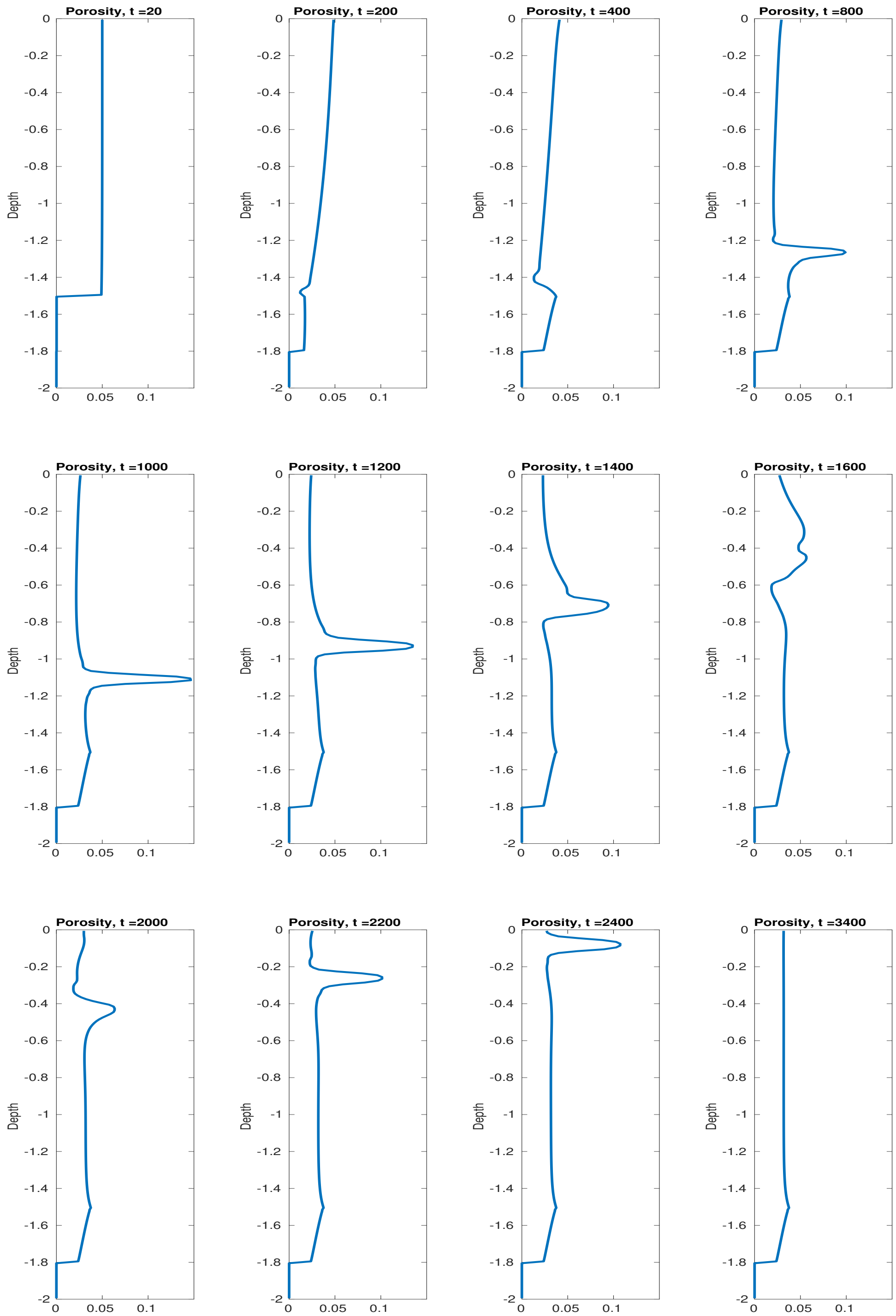

Figure 1.3: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of porosity shown at different time steps.

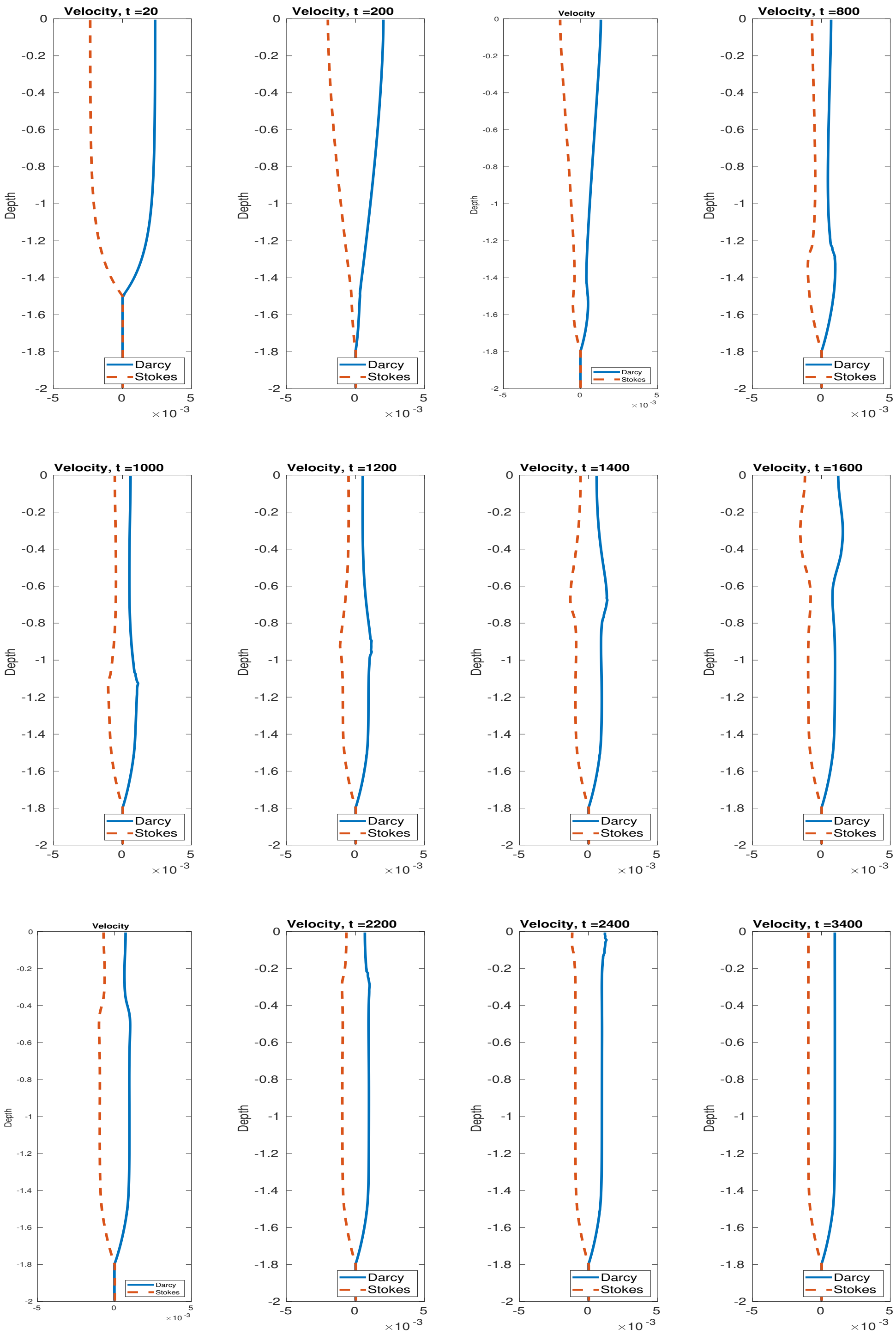

Figure 1.4: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of velocities shown at different time steps.

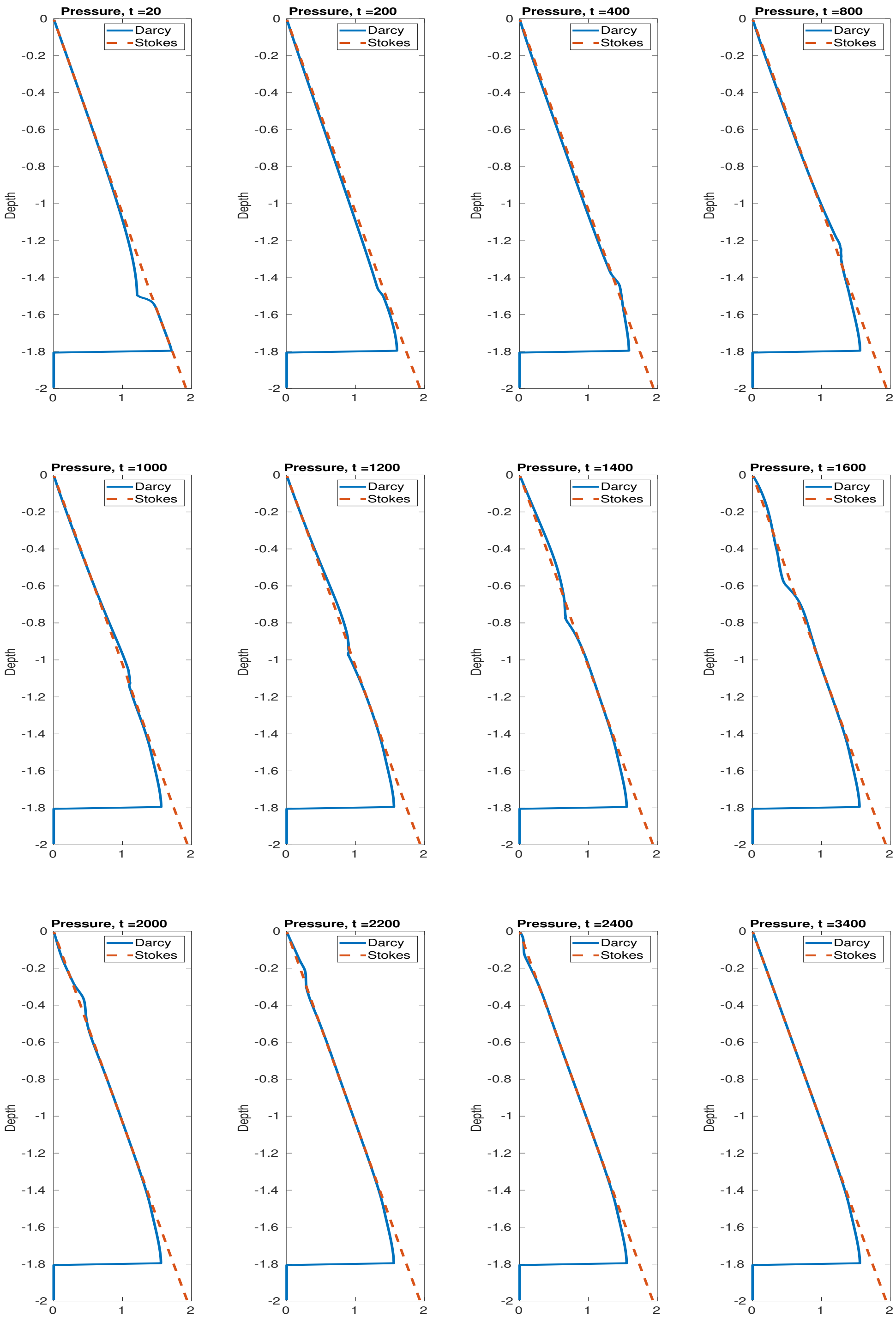

Figure 1.5: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of pressures shown at different time steps.

1.2.2 Phase Coupled Calculation

In the subsequent sections, we couple the MFEM and FVM solvers with the eutectic phase package to complete the fully integrated simulation framework. In this setting, porosity is no longer prescribed through an externally imposed melting function but is instead computed consistently from the transported nondimensional bulk composition and enthalpy. The computational domain represents an upwelling mantle column, with corner flow neglected near the surface under the simplifying assumption that the column continues its vertical ascent through the lithosphere. This configuration constitutes a minimal yet dynamically representative test case for assessing the stability and accuracy of the coupled algorithm in the presence of phase transitions and melt–solid interactions.

We then propose some simplifications to our simulations. First of all, we reconsider the transport equations (**??**) and the simplified version is shown as

$$\begin{aligned} \frac{\partial C}{\partial \check{t}} + \nabla \cdot \check{\mathbf{v}}_{\text{eff}} C = 0, \\ \frac{\partial H}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}}\check{T}) - \check{L}\nabla \cdot ((1-\phi)\check{\mathbf{v}}_s) - \frac{\alpha_\rho l_0}{c_p}\check{T}\check{\mathbf{v}} \cdot \mathbf{g} = 0. \end{aligned} \tag{1.8}$$

In this formulation, the diffusive contributions in both the mass composition and enthalpy equations are neglected. Specifically, the diffusion term $\frac{\phi \mathcal{D}_l}{u_0 l_0}\nabla c_l$ in the liquid phase is omitted due to the fact that the porosity ϕ remains small ($\phi \ll 1$) and the associated diffusivity coefficient $\mathcal{D}_l$ is comparatively negligible (on the order of 10^{-10} to 10^{-9} m^2/s) compared to effective advective velocities.

Likewise, while thermal conduction is an important mechanism in mantle convection over geological timescales, its influence on the present simulations is negligible. Two considerations motivate this simplification. First, the temperature profile maintains either constant or linear, such that the Laplacian of temperature vanishes or remains small throughout the evolution. Second, we do not impose a non-permeable solidification boundary at the surface, which would otherwise establish sharp thermal gradients and necessitate conductive heat transport across a boundary layer, in which case the diffusivity is no longer negligible (see Hewitt and Fowler (2008)).

Additionally, given that the primary focus of this study is the phenomenon of partial melting under small-porosity conditions, we adopt the lithostatic pressure as the reference pressure in determining the local melting point (see Katz (2008)). This approximation is justified because the dynamic pressure perturbations associated with porous flow are small compared to the background lithostatic gradient. As the mantle column ascends from depth, the concomitant reduction in lithostatic pressure leads to a decrease in the solidus temperature, thereby inducing some degrees of melting in proportion to the degree of decompression.

1.2.2.1 Boundary conditions

In this set of simulations, we do not attempt to prescribe fully physically consistent boundary conditions, owing to the inherent complexity of capturing both the mechanical and thermochemical constraints at the mantle–lithosphere interface. For example, Hewitt and Fowler (2008) formulated boundary conditions that simultaneously satisfy conservation laws and thermodynamic equilibrium at the surface. These constraints introduce additional complexity to model and numerical scheme that is not central to the phenomena under investigation here.

By prescribing essential boundary conditions on the bottom and the lateral edges and a natural (free-traction) boundary at the top for the Darcy–Stokes MFEM solver, the simulations allow for mechanically reasonable mantle upwelling without artificial confinement. Similarly, the Dirichlet inflow and free outflow conditions for the transport solver ensure a constant supply of chemical species and energy while permitting realistic advective transport, without imposing restrictive thermodynamic constraints. This approach maintains the internal consistency of the coupled MFEM–FVM solver and captures the key dynamics of melt migration, porosity evolution and phase transitions, while reducing computational complexity.

1.2.2.2 Numerical Flux

We modify the semi-discretized formulation of a general conservation law shown in equation (1.9) to include the "Source" term S as

$$\frac{\partial \overline{u}_E}{\partial t} + \sum_{i=0}^{4} \frac{|e_i|}{|E|} \sum_{g=0}^{G} \omega_{i,g} F(\mathcal{R}_{i,g}^{+}, \mathcal{R}_{i,g}^{-}) \cdot \boldsymbol{\nu}_i + \sum_{h=0}^{H} \xi_h S(\mathcal{R}_h) = 0, \tag{1.9}$$

where $\omega_{i,g}$ represents the weight associated to the g^{th} quadrature points on edge e_i and ξ_h represents the weight associated to the h^{th} quadrature points on face. In practice, we can simplify the numerical face integration by using mid-point rule and directly using the cell-averaged transport variable for second order accuracy.

The adiabatic cooling term is regarded as the "source" term and is straightforward to implement. We then write the corresponding Lax-Friedrich numerical flux of mass composition and enthalpy across the edges in terms of ML-WENO reconstructed values $\mathcal{R}_C$ and $\mathcal{R}_H$ as:

$$\begin{aligned} F_C(\mathcal{R}_C^{+}, \mathcal{R}_C^{-}) &= \frac{1}{2} \check{\mathbf{v}}_{\text{eff}} \cdot \boldsymbol{\nu} \Big[\mathcal{R}_C^{+} + \mathcal{R}_C^{-} - \alpha_C \big(\mathcal{R}_C^{+} - \mathcal{R}_C^{-} \big) \Big], \\ F_H(\mathcal{R}_H^{+}, \mathcal{R}_H^{-}) &= \frac{1}{2} \check{\bar{\mathbf{v}}} \cdot \boldsymbol{\nu} \Big[\check{T}^{+} + \check{T}^{-} - \alpha_T \big(\check{T}^{+} - \check{T}^{-} \big) \Big] - \check{L}(1-\phi) \check{\mathbf{v}}_s \cdot \boldsymbol{\nu}, \end{aligned} \tag{1.10}$$

where $\check{T}^{+} = \check{T}(\mathcal{R}_H^{+}, \mathcal{R}_C^{+})$ and $\check{T}^{-} = \check{T}(\mathcal{R}_H^{+}, \mathcal{R}_C^{+})$ are temperatures evaluated at both sides of the edge with phase package using the reconstructed pointwise mass composition and enthalpy values.

Following the disscussion of obatining a compatible porosity on the edge in section 1.1.1.

Some discussion on effective velocity $\mathbf{v}_{\text{eff}}$: We now

In the phase coupled calculation case, besides the values of porosity, the primary transport variables are also reconstructed differently due to no melt regions and distinct eutectic phase regions. We view it as a nonlinear transport problem and thus employ the local Lax-Friedrich stabilization scheme.

1.2.2.3 Two cases

In this section, We presenting two results with different domain lengths and initial conditions

Works Cited

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